

BRAC UNIVERSITY

Department of Computer Science and Engineering

Examination: Quiz - 1
Duration: 25 minutes

Semester: Spring 2026
Full Marks: 20

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID: _____	Section: _____
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1.

Instruction Set:

Add
Sub
Ld
Add
Sd
Sub
Sub
Sll
Add
Sub
Sub
Sub
Ld
Sd
Sub
Sll

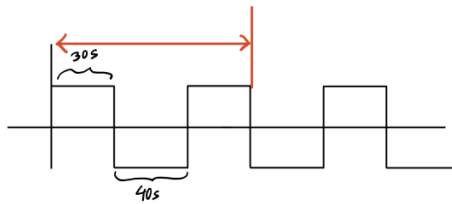


Figure : Add Instruction

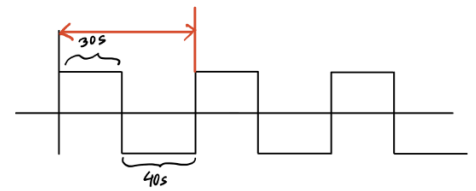


Figure : Sub Instruction

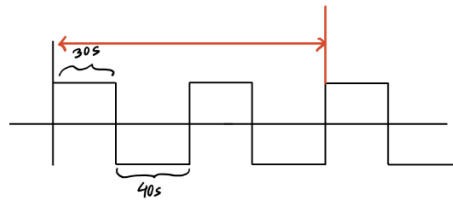


Figure : Ld Instruction

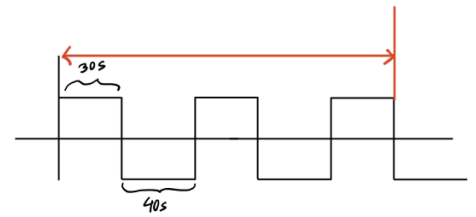


Figure : Sd Instruction

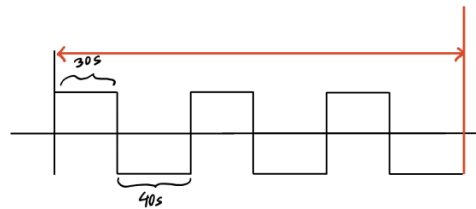


Figure : Sll Instruction

a. Find CPIs; You must answer in the following table. [1 x 5]

	Add	Sub	Ld	Sd	Sll
CPI	1.5	1	2	2.5	3
IC	3	7	2	2	2

b) Clock Cycle Time = (30 + 40) ns

= 70 ns

Freq. = $\frac{1}{\text{Clock Cycle Time}}$

= $\frac{1}{70}$ Hz

= 0.0142×10^{-9} GHz

b. Find the Frequency in GHz. Answer: _____ [1]

c. Find the time to run the instruction set. Answer: _____ [4]

c) Avg. CPI = $\frac{(1.5 \times 3) + (1 \times 7) + (2 \times 2) + (2.5 \times 2) + (3 \times 2)}{16} = \frac{26.5}{16} = 1.65$

CPU Time = $16 \times 1.65 \times 70 = 1855$

2. A processor **tries to guess the next instruction it will need to execute based on patterns in past instructions**. If the guess is correct, execution continues without waiting, improving overall speed.

Write the name of the relevant idea from the 7 Great Idea(s) that is applied in the given scenario. [2]

Answer: **Performance via Prediction**

3. Spec ratio of a program running on Computer A is 0.97. The reference computer has a 4 GHz clock. The total clock cycle needed to run the program on reference computer is 20×10^9 .

a. Find the execution time of the program in Computer A. [3]

b. Assuming both the reference computer and Computer A follow the same ISA, and each instruction takes 2 clock cycles to execute on both computers. Find the MIPS of Computer A. [4]

$$\begin{aligned}
 \text{a) Ref. time} &= \text{CPU time of ref. computer} \\
 &= \frac{\text{Clock Cycle}}{\text{Freq.}} \\
 &= \frac{20 \times 10^9}{4 \times 10^9} \\
 &= 5 \text{ s}
 \end{aligned}$$

$$\text{Spec Ratio} = \frac{\text{Ref. time}}{\text{CPU time}}$$

$$\Rightarrow \text{CPU time} = \frac{\text{Ref. time}}{\text{Spec Ratio}} = \frac{5}{0.97} = 5.15 \text{ s}$$

$$\begin{aligned}
 SR &= 0.97 \\
 F_{\text{ref}} &= 4 \text{ GHz} \\
 &= 4 \times 10^9 \text{ Hz} \\
 \text{C.C.}_{\text{ref}} &= 20 \times 10^9
 \end{aligned}$$

b) Same ISA & Same Prog. $\rightarrow$ Inst. Count same.

CPI also same on both computers.

Clock Cycle = IC $\times$ CPI ; $\rightarrow$ same on both computers.

$$\begin{aligned}
 \text{IC} &= \frac{\text{Clock Cycle}}{\text{CPI}} \\
 &= \frac{20 \times 10^9}{2} = 10 \times 10^9 \rightarrow 5.15 \text{ s}
 \end{aligned}$$

$$\begin{aligned}
 \text{Mips} &= \frac{\text{IC}}{\text{CPU Time} \times 10^6} \\
 &= \frac{10 \times 10^9}{5.15 \times 10^6} \\
 &= 1941.74
 \end{aligned}$$